Yuyang (Daniel) Shan

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

B.S. in Music and Technology (Computer Science Track), Minor in Artificial Intelligence

B.F.A. in Music Composition

Expected May 2027 | GPA: 3.86/4.00

RESEARCH EXPERIENCE

Independent Research (Capstone Project): Automatic Piano Fingering Estimation

Carnegie Mellon University | Advisor: Chris Donahue | 2026 – Present

- Built and systematically benchmarked five model families (feature bigram, HMM, CRF, Transformer encoder, and encoder-CRF) under a shared multi-ground-truth evaluation protocol; identified limited generalizable training signal as the primary performance bottleneck rather than model capacity or modeling paradigm.
- Achieved 0.594 held-out test M_{gen} with a preregistered neural candidate, surpassing the strongest hand-crafted CRF baseline (0.585) for the first time in the project under evaluation rules fixed before a one-time test-set evaluation.
- Engineered a ~4.4K-score pretraining corpus from PDMX, including n-gram leakage checks and content-level deduplication that removed ~31% metadata-invisible duplicates; designed masked score-fingering pretraining and observed +2.1 M_{gen} points for each 3.7× increase in corpus size.
- Found that fixed-epoch budgets systematically biased objective ablations against slower-converging multi-objective models; retraining to convergence produced the project's largest frozen-probe gains (+2.9 to +3.6 points) and overturned three earlier experimental conclusions.
- Extended fingering estimation to polyphonic music; achieved 0.717 held-out test M_{gen} with validation-selected local attention, outperforming the full-attention counterpart (0.700). Found that wider context improved score reconstruction but not downstream fingering prediction.

Research Assistant: Roger Dannenberg

Carnegie Mellon University | 2026 – Present

- Built a pipeline to generate draft performance-to-score alignments and automatically flag likely errors; manually corrected each piece by ear and produced an ASAP-style aligned dataset for the opera *Felicità*.
- Compared beat-interpolated ASAP with nASAP note-level alignments across 1,065 performances to address a reviewer-raised concern about dataset choice; quantified their agreement (10 ms median), verified nASAP defects, and combined score inspection with listening to distinguish alignment errors from notation-inherent ambiguity.
- Identified and corrected a bug in an existing score-following evaluation reader that included non-event rows, substantially revising the reported error distributions and the HMM versus Accomplice comparison.
- Developed rule-based online performance-time prediction methods that respect causal timing constraints and previously committed predictions, with onset-based evaluation against HMM and Accomplice.
- Contributed a structured score-following literature review to a group paper in preparation.

SELECTED TECHNICAL PROJECTS

Independent Project: SATB Harmony Grader & Solver

- Implemented a rule-based SATB harmony grader by adapting, with permission, Dr. John Paul Ito's harmony curriculum into 11 categories of diagnostic feedback covering voice leading, spacing/range, chord completion, parallel/direct intervals, leaps, and tendency-tone/chromatic-chord resolution.
- Designed a two-stage heuristic SATB solver that searches rule-valid realizations while optimizing musical preferences including smooth voice leading, comfortable ranges and spacing, cadence-aware soprano planning, fixed-note attraction, and phrase-level melodic shaping.
- Built separate Flask-based teacher and student interfaces with interactive VexFlow notation, SoundFont playback, JSON assignment/submission exchange, optional student self-checking, and one-click grading/reference solutions.

TECHNICAL SKILLS

Programming: Python, C, Standard ML

Machine Learning / Scientific Computing: PyTorch, NumPy

Music Computing & Data: MIDI, MusicXML, Nyquist

HONORS & AWARDS

Finalist (Second Place), 2026 Philharmonic Soloist Competition, Carnegie Mellon University School of Music.

Presser Undergraduate Scholar Award, Nominee, Carnegie Mellon University School of Music (2026–27; decision pending).

Dean's List, Carnegie Mellon University: Fall 2024, Spring 2025, Fall 2025.

MUSIC BACKGROUND

Composition: Orchestral, chamber, vocal, and solo works; principal teacher: James Whipple

- Piano Concerto in C-sharp minor (2025–26), for piano and string quintet, three movements, ca. 25 min.
- Piano Scherzo in G, the 2026–27 Senior Orchestral Project; premiere by the CMU Philharmonic scheduled for March 21, 2027.
- Viola Concertino in B minor (in progress), for viola and wind quintet, three parts, ca. 12 min.

Piano: Classical pianist working in solo, chamber, and concerto repertoire; principal teacher: Sung-Im Kim

- Composer-pianist in premieres of own works, including the Piano Concerto in C-sharp minor at CMU Composer Concerts, 2025–26.
- Ravel, Piano Concerto in G major, complete, in the final round of the 2026 CMU Philharmonic Soloist Competition.
- Schubert, Piano Quintet in A major, D. 667 ("Trout"), 2026.
- Premieres of new works by fellow CMU composition students, 2024 and 2025.